Roll No. .....

TEC–404

B. TECH. (ECE) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

ELECTROMAGNETIC FIELD THEORY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Give vector $\vec{A} = 3\hat{a}_x - 2\hat{a}_y + \hat{a}_z$ and $\vec{B} = \hat{a}_x - 3\hat{a}_y + 2\hat{a}_z$. Determine the Determine the vector projection of $\vec{A}$ on $\vec{B}$ and unit vector perpendicular to both $\vec{A}$ and $\vec{B}$. (CO1)

OR

(b) Determine the scalar triple product, vector triple product and angle between $\vec{P}$ and $\vec{Q}$ for the following vectors : (CO1)

$$\vec{P} = 2\hat{a}_x + 3\hat{a}_y - \hat{a}_z$$

$$\vec{Q} = \hat{a}_x - \hat{a}_y - \hat{a}_z$$

$$\vec{R} = 3\hat{a}_x - 4\hat{a}_y - 2\hat{a}_z.$$

P. T. O.

2. (a) Express the following vectors in Cartesian coordinates : (CO1)

$$\vec{A} = r^2 \hat{a}_r + \sin\theta \hat{a}_\phi$$

OR

(b) Give point P (– 2, 6, 3) and vector $\vec{A} = y\hat{a}_x + (x+z)\hat{a}_y$. Express P and $\vec{A}$ in spherical coordinate. Evaluate $\vec{A}$ at P in spherical systems. (CO1)

3. (a) Express the following vectors is Cartesian coordinates : (CO1, CO2)

$$\vec{C} = z\sin\phi \hat{a}_\rho - \rho\cos\phi \hat{a}_\phi + 2\rho z \hat{a}_z$$

OR

(b) Determine the gradient of the following scalar field : (CO1, CO2)

$$f = \cos\theta \sin\phi \ln r + r^2 \phi$$

4. (a) Determine the divergence of the following vector fields and evaluate them at the specified points : (CO2)

$$\vec{B} = \rho z \sin\phi \hat{a}_\rho + 3\rho z^2 \cos\phi \hat{a}_\phi \text{ at } (5, \pi/2, 1)$$

OR

(b) Determine the curl of the following vector fields and evaluate it at the specified points : (CO2)

$$\vec{A} = yz\hat{a}_x + 4xy\hat{a}_y + y\hat{a}_z \text{ at } (1, -2, 3)$$

5. (a) What do you understand by the Gauss Law ? Determine the electric field intensity due to finite surface charge and infinite line charge with help of Gauss law. (CO2)

OR

(b) If :

$$\vec{D} = \left(2y^2 + z\right)\hat{a}_x + 4xy\hat{a}_y + x\hat{a}_z \text{C/m}^2 .$$

find : (CO2)

(i) the volume charge density at (–1, 0, 3)

(ii) Total charge enclosed by the cube $0 \le x \le 1, 0 \le y \le 1, 0 \le z \le 1$.

Roll No. .....

TEC–401

B. TECH. (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

COMMUNICATION SYSTEMS–I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) (i) Derive an expression for a single tone amplitude modulated wave.

(ii) An unmodulated carrier has a peak value of V volt. When it is amplitude modulated with a sine wave, the maximum peak to peak value (of the modulated carrier) is 3 V volt. Determine the modulation index. (CO1)

OR

(b) (i) Calculate the percentage power saving when the carrier and one of the sidebands are suppressed in an AM wave modulated to a depth of (a) 100 percent and (b) 50 percent.

(ii) An AM modulator has output $x(t) = A \cos 400\pi t + B \cos 380\pi t + B \cos 420\pi t$. The carrier power is 100 W and the efficiency is 40%. Determine the value of A and B. (CO1)

P. T. O.

2. (a) Explain the need for modulation in a communication system. (CO1)

OR

(b) Explain the operation of the Ring modulator circuit which generate the DSB-SC wave. (CO1)

3. (a) (i) Show that a signal and its Hilbert transform are orthogonal.

(ii) Show that a signal and its Hilbert transform have the same energy content. (CO1)

OR

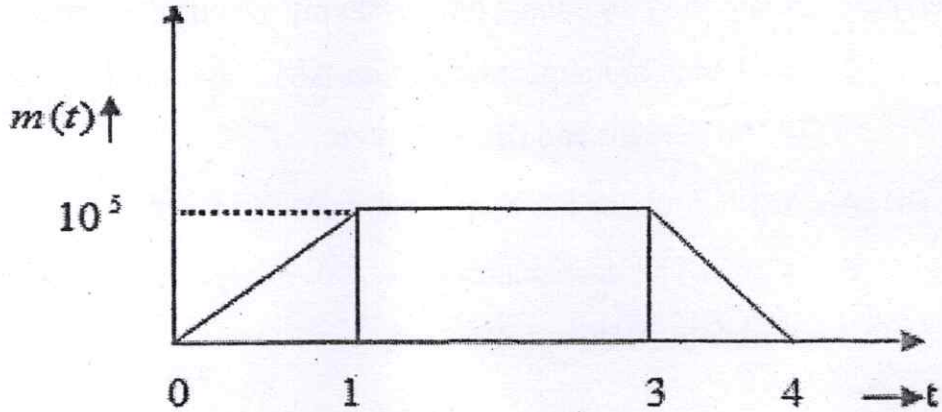
(b) Explain demodulation of DSB-SC wave using synchronous detector. Also, discuss the frequency and phase errors in synchronous detector. (CO1)

4. (a) Explain envelope detector method for demodulator of AM waves. Also, derive the relation for obtaining optimum value of time constant RC. (CO1)

OR

(b) Show that DSB-SC amplitude modulation is linear while angle modulation is not. (CO1)

5. (a) For the message signal $m(t)$ shown in figure below : (CO2)



(i) If $m(t)$ is frequency modulated on a carrier with frequency 10^6 Hz, with a frequency deviation constant equal 5 Hz/volt, what is the maximum instantaneous frequency of the modulated signal ?

(ii) If $m(t)$ is phase modulated on a carrier with frequency 10^6 Hz, with a phase deviation constant equal to 3 rad/volt, what is the maximum instantaneous frequency of the modulated signal ?

OR

(b) Derive an expression for a single tone frequency modulated wave.

(CO2)

60

Roll No. .....

TEC–403

B. TECH. (ECE) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

MICROPROCESSOR AND ITS APPLICATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Differentiate between the following instructions with suitable example : (CO1)

   (i) LDA and LDAX

   (ii) LXI and MVI

OR

   (b) Write a program to find number of ones in an 8 -bit data stored at location 2050 store result at 3020. (CO1)

2. (a) Draw the timing diagram for instruction SUB M. (CO1, CO2)

OR

   (b) What is Interrupt ? Explain Hardware interrupts in 8085. (CO1, CO2)

3. (a) What is addressing modes ? Explain addressing modes of 8085 with example. (CO2)

P. T. O.

OR

(b) Write a program for division of two 8-bit numbers stored at location 2050 and 2051 store result at 3050. (CO2)

4. (a) Explain interfacing of 32 kB memory chips with 8085 with suitable diagram. (CO1)

OR

(b) Write a program to find factorial of a number stored at location 2050 store result at 2020. (CO1)

5. (a) Explain programming model of 8085. (CO2)

OR

(b) Write a program to transfer 25 bytes of data from location 2020 to 3020. (CO2)

60